ATTENTION AI

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CERTIFICATE

This is to certify that research work embodied in this report entitled Attention AI was carried out by **Sarna Karandeep Singh 202126940109** at the Centre for Professional Course for partial fulfilment of M.Sc. IT degree to be awarded by Gujarat University. This research work has been carried out under my supervision and is to the satisfaction of the department.

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DECLARATION OF ORIGINALITY

I hereby certify that I am the sole author of this Project report and that neither any part of this Project report nor the whole of the Project report has been submitted for a degree to any other University or Institution.

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PROJECT REPORT APPROVAL

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Chapter 1 — Introduction

1.1 Introduction

- **Attention** plays a very important role in activities like learning, working, and performing tasks that require focus. When a person is not attentive, their performance decreases, mistakes increase, and overall efficiency becomes low, especially in environments like online education or remote work.
- In **online environments**, monitoring attention becomes difficult because there is no physical presence of a teacher or supervisor to observe behaviour continuously. This creates a strong need for an automated system that can track attention without human involvement.
- This project develops an **AI-based attention monitoring system** that uses a webcam to detect whether a person is attentive or distracted in real-time. The system continuously analyses visual input and provides instant feedback.
- The system uses **computer vision techniques** to detect facial features such as eyes, face position, and head movement, which are important indicators of attention and focus.
- A simple and user-friendly **GUI interface** is designed so that the user can easily start the system, view results, and understand the attention status without any technical difficulty.

1.1.1 Overview of Attention Monitoring Systems

- **Attention monitoring systems** are systems designed to track and measure whether a person is focused on a task or not. These systems mainly use camera input or sensors to observe behaviour such as eye movement, facial expressions, and head direction.
- Traditional systems rely on **manual observation**, where a teacher or supervisor watches the user, but this method is not efficient, cannot be done continuously, and is prone to human error.
- Modern systems use **Artificial Intelligence (AI)** to automate the process, making it faster, more accurate, and capable of working continuously without fatigue or mistakes.

- These systems analyse multiple features like **eye gaze, face orientation, and head pose**, which together provide a better understanding of the user's attention level.
- Example: In **online examination systems**, attention monitoring is used to detect suspicious behaviour like looking away from the screen or being distracted frequently.

Diagram 1 — Overview of attention monitoring systems

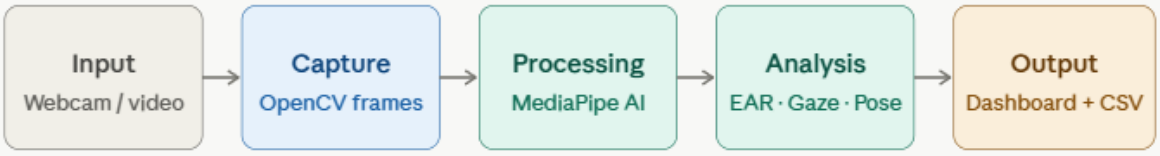


Fig 1 — Overview of attention monitoring system pipeline

1.1.2 Importance of AI in Real-Time Monitoring

- **Artificial Intelligence (AI)** enables systems to process large amounts of visual data automatically and make decisions based on patterns without requiring human intervention. This makes monitoring more efficient and scalable.
- **Real-time monitoring** means the system can detect and respond instantly while the user is being observed, which is very important in applications like online classes or safety systems.
- AI models are capable of analysing complex patterns such as **eye movement, blinking, and head direction**, which are difficult to measure manually with accuracy.
- Using AI improves both the **speed and accuracy of detection**, ensuring that the system gives reliable results in different conditions.
- It also allows the system to be used in various real-world applications like education, workplace monitoring, and driver safety systems.

1.2 Motivation

- In online learning environments, students often **lose focus easily**, which negatively affects their understanding and performance. This

problem becomes more serious when there is no monitoring system in place.

- Teachers and instructors cannot observe every student continuously in a virtual classroom, which makes it difficult to identify who is paying attention and who is distracted.
- There is a strong need for a **simple, automated, and real-time solution** that can monitor attention without requiring manual supervision.
- AI-based systems can help improve **productivity, engagement, and learning outcomes** by providing continuous monitoring and feedback.
- This project is motivated by the idea of using **basic hardware like a webcam and simple AI techniques** to create an effective and practical solution for real-world problems.

1.3 Objectives

- The main objective of this project is to develop a **real-time attention detection system** using Artificial Intelligence and computer vision techniques that can monitor user behaviour continuously.
- The system should be able to detect **face, eyes, and head movements** accurately using webcam input and analyse them to determine attention level.
- It should classify whether the user is **attentive or not attentive** based on predefined conditions and visual indicators.
- A simple and easy-to-use **Graphical User Interface (GUI)** should be designed so that users can interact with the system without any difficulty.
- The system should be efficient, lightweight, and capable of running on **normal computers without requiring high-end hardware.**

1.4 Organization of Report

- Chapter 1 provides an introduction to the project, including the motivation, objectives, and overall idea behind developing the system. It sets the context for understanding the problem.
- Chapter 2 explains the background theory related to computer vision, machine learning, and attention detection concepts, which are necessary to understand how the system works.

- Chapter 3 presents the literature survey, where existing research and systems are discussed and compared to identify gaps and improvements.
- Chapter 4 describes the proposed system, including the architecture, methodology, and approach used to solve the problem.
- Chapter 5 to Chapter 8 cover implementation details, results, performance evaluation, conclusion, and future scope of the project.

Chapter 2 — Background Theory

2.1 Computer Vision Basics

- **Computer Vision** is a branch of Artificial Intelligence that enables machines to interpret and understand visual information from images and videos. It allows systems to detect objects, faces, and movements automatically.
- In this project, computer vision is used to process **real-time webcam video** and analyse the user's facial features continuously to determine their attention level.
- It helps in identifying important features such as **face position, eye direction, and head movement**, which are essential for attention detection.
- Libraries like **OpenCV and MediaPipe** provide powerful tools and pre-built models that make it easier to implement computer vision applications.
- This concept forms the **foundation of the entire system**, as all detection and analysis depend on visual data processing.

Diagram 2 — Computer vision basics: camera to processing pipeline

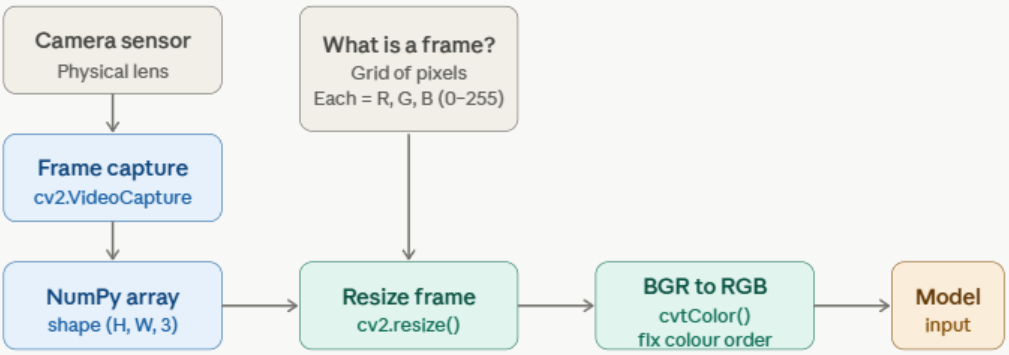


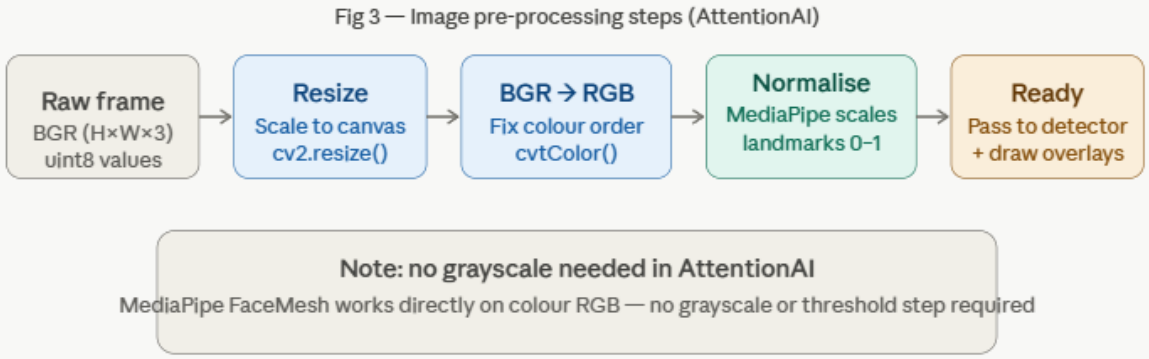
Fig 2 — Computer vision basics: camera to processing pipeline

2.1.1 Image Processing

- **Image processing** is the technique of modifying and analysing images to extract meaningful information. It involves operations like resizing, filtering, enhancing, and converting image formats.

- In this project, each frame captured from the webcam is processed before analysis to ensure better **accuracy and clarity of detection.**
- Converting images to formats like **RGB** helps models understand the image data more effectively and improves performance.
- Continuous processing of frames allows the system to work in **real-time without delays**, which is important for live monitoring.
- This step acts as a **preprocessing stage** before applying face detection and landmark detection algorithms.

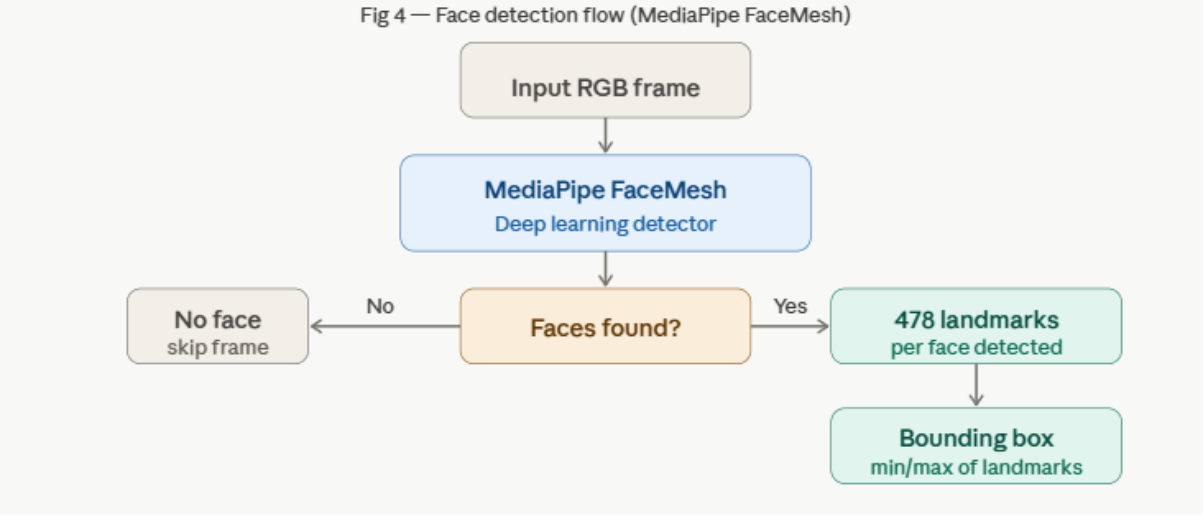
Diagram 3 — Image pre-processing steps



2.1.2 Face Detection

- **Face detection** is the process of identifying whether a human face is present in an image or video frame and locating its position. It is the first step in the attention detection system.
- The system uses **MediaPipe's face detection model**, which is fast, efficient, and suitable for real-time applications.
- If a face is detected, the system proceeds to further analysis; otherwise, it assumes that the user is **not present or not attentive**.
- Accurate face detection ensures that the system only processes relevant frames and avoids unnecessary computation.
- This step is essential because all further operations like eye tracking and head pose estimation depend on detecting the face first.

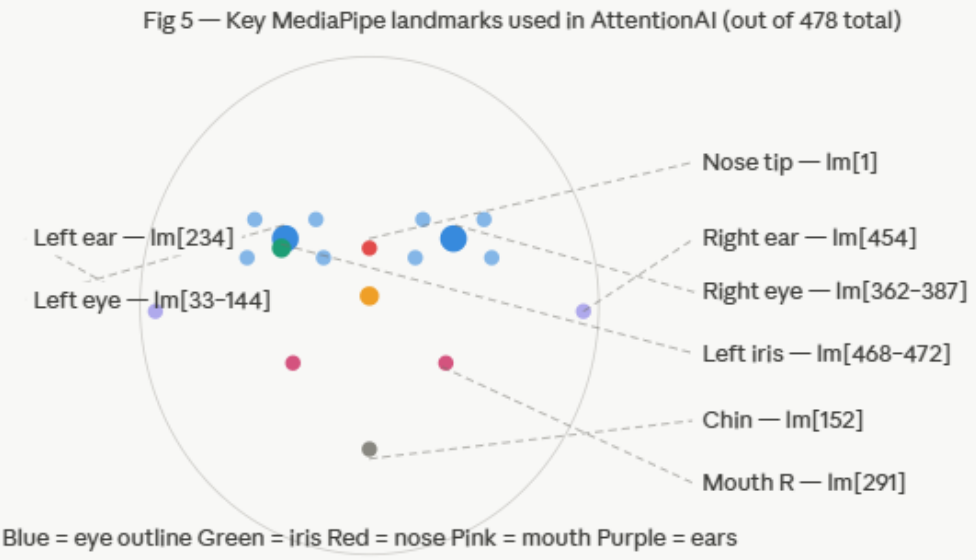
Diagram 4 — Face detection flow



2.1.3 Facial Landmark Detection

- **Facial landmark detection** involves identifying key points on the face such as eyes, nose, mouth, and jawline, which help in tracking facial movements precisely.
- MediaPipe provides around **468 landmark points**, which give a detailed map of the face and allow accurate tracking of small movements.
- These landmarks are used to detect **eye position, blinking, and head orientation,** which are important for determining attention.
- By analysing these points, the system can understand whether the user is looking at the screen or looking away.
- This technique is a **core component** of the attention detection system and directly affects its accuracy

Diagram 5 — Facial landmark detection points



2.2 Machine Learning & Deep Learning

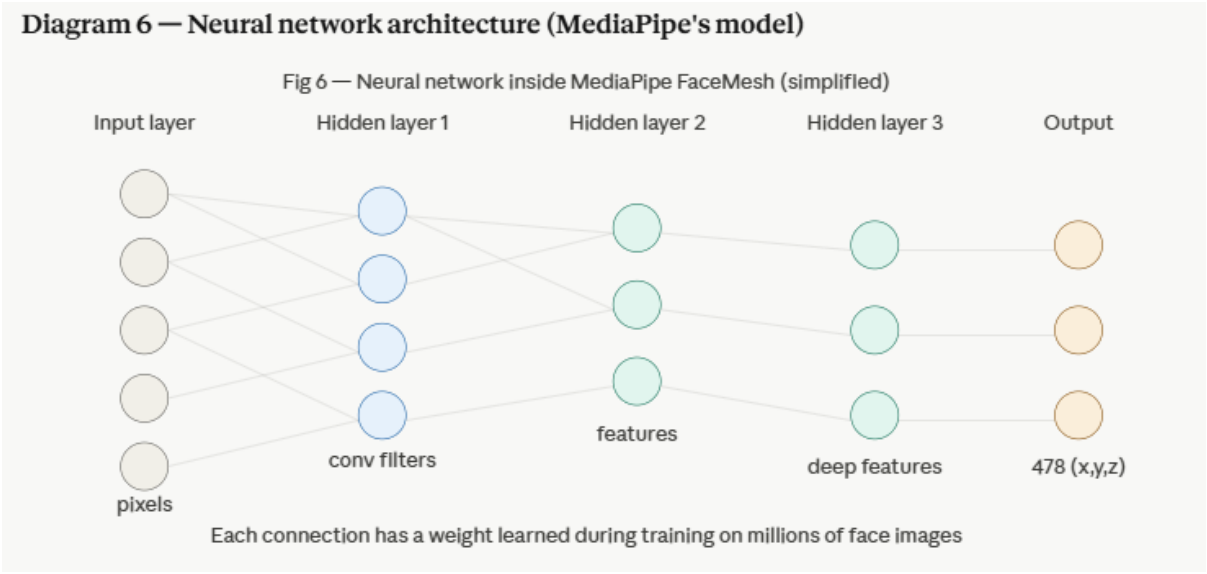
- **Machine Learning (ML)** is a subset of Artificial Intelligence that allows systems to learn patterns from data and make decisions without being explicitly programmed for every scenario. It improves performance over time by learning from examples.
- **Deep Learning (DL)** is a more advanced form of ML that uses multi-layered neural networks to learn complex patterns, especially in images, videos, and speech data. It is highly effective for tasks like face detection and recognition.
- In this project, ML and DL techniques are used to analyse **visual data from webcam frames** and identify patterns related to attention such as eye movement and head direction.
- These techniques help the system make **automatic and accurate decisions** about whether a user is attentive or not, without manual intervention.
- ML and DL form the **core intelligence** of the system, enabling real-time analysis and prediction.

2.2.1 Neural Networks

- **Neural Networks** are computational models inspired by the human brain, consisting of layers of interconnected nodes (neurons) that process information step-by-step.

- Each neuron receives input, processes it using weights and activation functions, and passes the output to the next layer, enabling the network to learn complex relationships.
- Deep learning models use **multiple hidden layers**, allowing them to detect patterns like facial features, eye positions, and head orientation with high accuracy.
- In this project, neural network-based models (like those used in MediaPipe) help in **facial landmark detection and feature extraction**.
- These networks improve the system's ability to handle variations such as different faces, lighting conditions, and angles.

Diagram 6 — Neural network architecture (MediaPipe's model)



2.3 Attention Detection Concepts

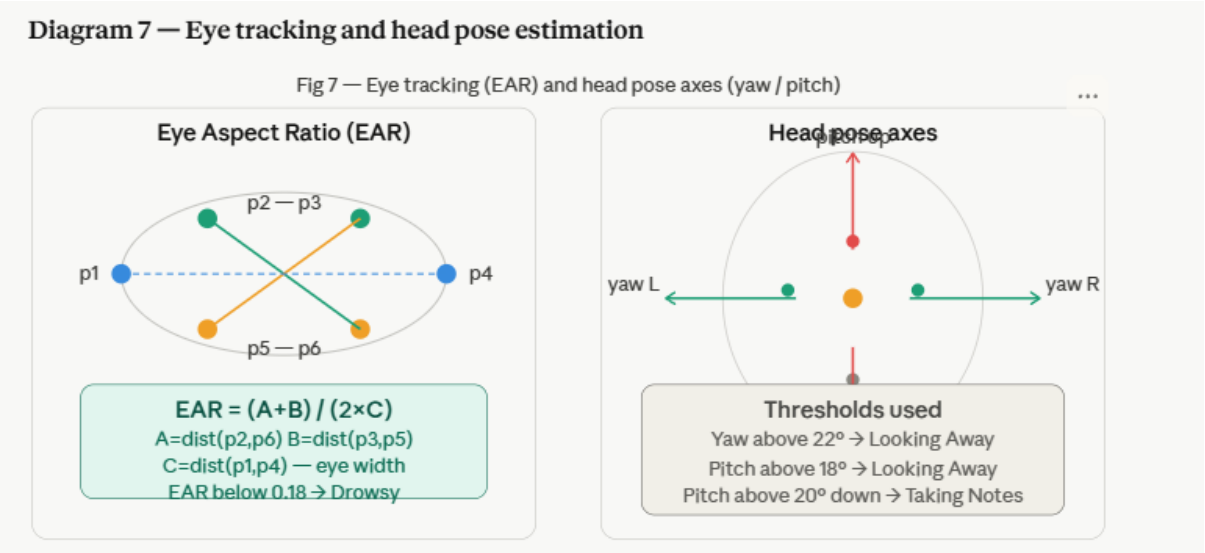
- **Attention detection** involves identifying whether a person is focused on a task by analysing their visual behaviour and movements.
- Key indicators of attention include **eye gaze direction, blinking rate, and head orientation**, which together provide a reliable measure of focus.
- If a user frequently looks away from the screen or their face is not aligned properly, the system classifies them as **inattentive**.
- The system continuously monitors these features in **real-time**, ensuring that attention status is updated instantly.

- This concept is essential for applications like **online learning, exams, and productivity monitoring**, where focus plays a critical role.

2.3.1 Eye Tracking & Head Pose Estimation

- **Eye Tracking** is used to determine where a person is looking by analysing the position and movement of the eyes. It helps in identifying whether the user is focused on the screen or distracted.
- The system uses **facial landmarks around the eyes** to estimate gaze direction and detect patterns like blinking or looking away.
- **Head Pose Estimation** determines the orientation of the head (left, right, up, down) using facial landmark positions and geometric calculations.
- By combining eye tracking and head pose, the system gets a **more accurate understanding of attention**, reducing errors that may occur if only one feature is used.
- These techniques are critical for ensuring **reliable and precise attention detection** in real-time applications.

Diagram 7 — Eye tracking and head pose estimation



Chapter 3 — Literature Survey

3.1 Literature Survey

- The **literature survey** focuses on studying existing research and systems related to attention monitoring, computer vision, and AI-based detection techniques. It helps in understanding how similar problems have been approached and solved.
- Many research works use **computer vision and machine learning** to detect user attention through facial features, eye movement, and head orientation. These methods are widely used due to their non-intrusive nature.
- Some systems on **eye-tracking hardware**, which provides high accuracy but increases cost and complexity, making them less practical for general users.
- Other approaches use **deep learning models** such as convolutional neural networks (CNNs) to improve detection accuracy, especially in complex conditions like low lighting or varied angles.
- Studies show that combining multiple features like **eye gaze, blinking rate, and head pose** results in better attention detection compared to using a single parameter.
- The survey highlights that while many solutions exist, there is still a need for a system that is **simple, cost-effective, and capable of real-time performance** on standard devices.

3.2 Comparative Study

- The **comparative study** analyses different existing systems based on factors like accuracy, cost, hardware requirements, and real-time capability to identify their strengths and weaknesses.
- Systems using **specialized hardware** provide high precision but are expensive and not easily accessible, limiting their use in common applications.
- Software-based systems using **AI and computer vision** are more flexible and affordable, but their performance can vary depending on environmental conditions.
- Some models focus on **high accuracy** but require heavy computation, while others are lightweight but may compromise slightly on precision.

- The proposed system achieves a balance by using **efficient algorithms and lightweight tools**, ensuring both good accuracy and real-time performance.
- Compared to existing approaches, this system is more **practical, user-friendly, and deployable on normal computers**, making it suitable for real-world usage.

Chapter 4 — Proposed Work

4.1 Problem Statement

- In online environments such as virtual classes and remote work, it is difficult to **monitor user attention continuously**, as there is no physical supervision present. This leads to reduced focus and lower productivity.
- Traditional methods of monitoring attention are either **manual, inefficient, or require expensive hardware**, making them unsuitable for large-scale or everyday use.
- There is a lack of systems that can provide **real-time, automated, and accurate attention detection** using simple and easily available resources.
- The problem is to design a system that can **detect whether a user is attentive or not** using only a webcam and software-based techniques.
- The solution should be **cost-effective, easy to use, and capable of running on normal computers** without requiring specialized equipment.

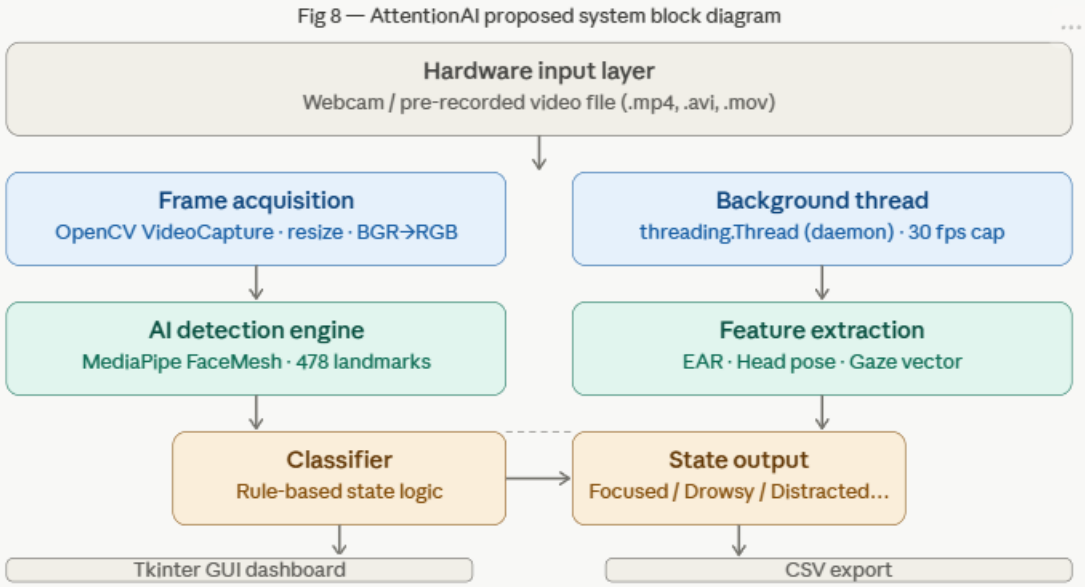
4.2 Research Gap

- Existing systems either focus on **high accuracy using complex models** or on simplicity with limited functionality, but very few provide a balance between both.
- Many solutions depend on **specialized hardware like eye trackers**, which increases cost and limits accessibility for general users.
- Some AI-based systems are not optimized for **real-time performance**, making them less effective for live monitoring applications.
- There is limited integration of **simple GUI-based interfaces**, which reduces usability for non-technical users.
- The gap lies in developing a system that is **real-time, accurate, lightweight, cost-effective, and user-friendly** at the same time.

4.3 Proposed System

- The proposed system is an **AI-based attention monitoring system** that uses a webcam to analyse user behaviour in real-time.
- It detects **facial features such as eyes and head position** to determine whether the user is focused on the screen.
- The system classifies the user as **attentive or inattentive** based on predefined conditions and continuously updates the result.
- A simple **Graphical User Interface (GUI)** is provided to allow easy interaction and display results clearly.
- The system is designed to be **lightweight and efficient**, ensuring smooth performance on standard devices.

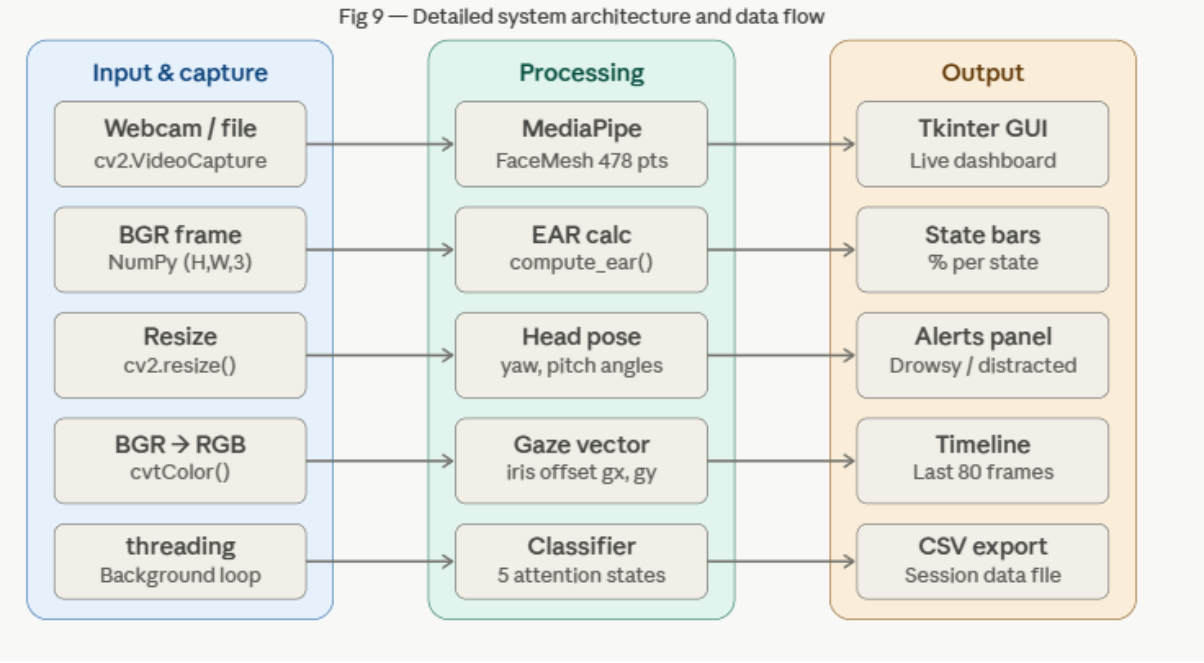
Diagram 8 — Proposed system block diagram



4.4 System Architecture

- The architecture is divided into key components: **input, processing, decision-making, and output**, ensuring a structured workflow.
- The **input module** captures real-time video using a webcam, which serves as the primary data source.
- The **processing module** performs face detection, landmark extraction, and feature analysis using computer vision techniques.
- The **decision module** evaluates the processed data to determine the attention level based on conditions like eye direction and head pose.
- The **output module** displays the result through a GUI, providing real-time feedback to the user.

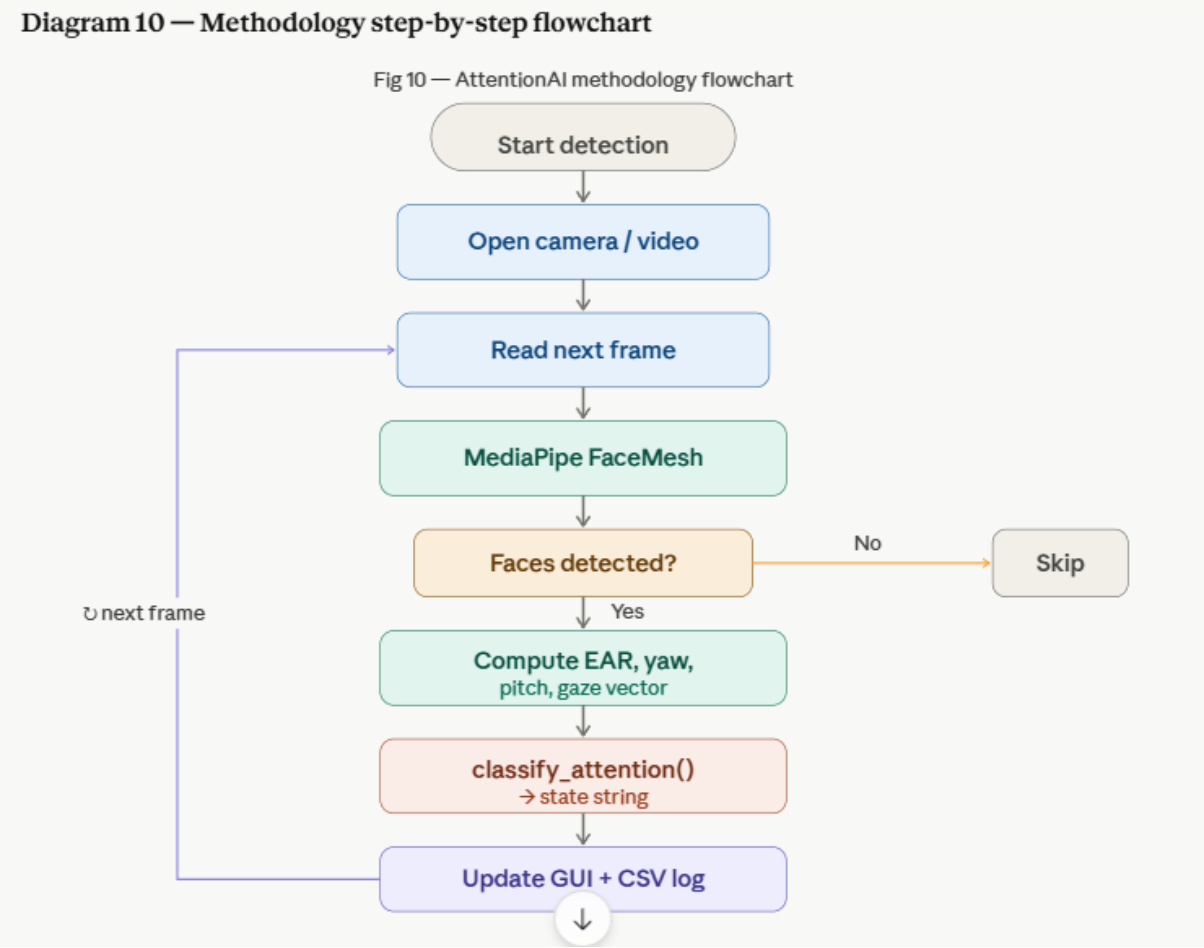
Diagram 9 — System architecture with detailed data flow



4.5 Methodology

- The system begins by capturing **continuous video frames** from the webcam to ensure real-time monitoring.
- Each frame is processed to detect the face and extract **facial landmarks** for detailed analysis.
- The system analyses **eye movement and head orientation** to identify whether the user is looking at the screen or distracted.
- Based on predefined rules, the system determines the **attention status** and updates the output instantly.
- This process runs in a loop, ensuring **continuous tracking and real-time feedback** without interruption.

Diagram 10 — Methodology step-by-step flowchart



4.6 Expected Outcome

- The system is expected to provide **accurate real-time detection** of user attention using only a webcam and basic computing resources.
- It will display clear results indicating whether the user is **attentive or inattentive**, improving awareness and focus.
- The solution will be **cost-effective and easy to use**, making it suitable for students, teachers, and professionals.
- It will demonstrate the practical application of **AI and computer vision** in solving real-world problems.
- The project will serve as a **foundation for future improvements**, such as integrating advanced models and additional features.

Chapter 5 — Implementation Environment

5.1 System Specifications

- The system is designed to run on **standard hardware**, ensuring accessibility without requiring high-end machines or specialized equipment. A basic laptop or desktop with moderate specifications is sufficient.
- Minimum requirements include a **processor (Intel i5 or equivalent), 8GB RAM, and a webcam**, which is essential for capturing real-time video input.
- The system requires sufficient storage space to install **required libraries and dependencies**, though it does not demand large disk capacity.
- A stable operating system such as **Windows, Linux, or macOS** is needed to support the development and execution environment.
- These specifications ensure that the system remains **lightweight, efficient, and widely usable** across different devices.

5.2 Programming Environment

- The project is developed using **Python**, which is widely used for Artificial Intelligence, Machine Learning, and computer vision applications due to its simplicity and strong library support.
- Development is carried out in environments like **Jupyter Notebook or Visual Studio Code**, which provide an interactive interface for coding, testing, and debugging.
- The system uses **Python libraries** for handling image processing, model execution, and GUI development, making the implementation more efficient.
- Required libraries are installed using package managers like **pip**, ensuring easy setup and maintenance of the development environment.
- The programming environment is chosen to ensure **ease of development, flexibility, and scalability**.

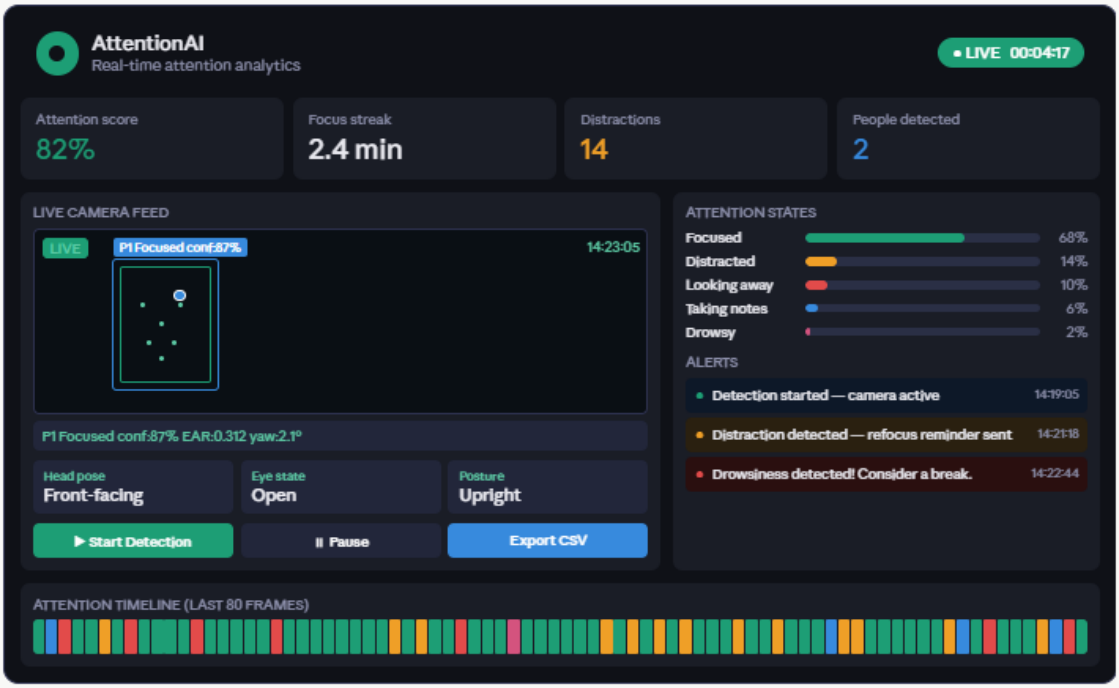
5.3 Tools and Technologies

- **OpenCV** is used for real-time image and video processing, enabling the system to capture frames from the webcam and perform operations like face detection and frame analysis.
- **MediaPipe** is used for facial landmark detection, providing accurate and efficient tracking of facial features such as eyes and head position.
- **Tkinter** is used to develop the Graphical User Interface (GUI), allowing users to interact with the system easily and view results in real-time.
- **NumPy** is used for numerical operations and handling arrays, which are essential for processing image data efficiently.
- These tools and technologies work together to create a system that is **fast, accurate, and capable of real-time performance** while remaining simple to implement.

Chapter 6 — Implementation Details

Diagram 11 — UI mockup / implementation screenshot

Fig 11 — AttentionAI UI implementation (representative mockup)



6.1 Data Input (Webcam Feed)

- The system captures **real-time video input** using a webcam, which serves as the primary data source for attention detection. Continuous frame capture ensures that the system can monitor user behaviour without interruption.
- Each frame from the webcam is read sequentially and converted into a suitable format for processing, allowing further analysis using computer vision techniques.
- The video stream is processed at a consistent frame rate to maintain **smooth and real-time performance**, avoiding delays or lag in detection.
- Proper handling of input ensures that frames are **efficiently transferred to the processing module**, reducing computational overhead.

- This stage forms the **foundation of the system**, as all further operations depend on the quality and consistency of the captured video.

6.2 Face & Landmark Detection

- The system first performs **face detection** to identify whether a human face is present in the frame and to locate its position accurately. This step ensures that only relevant data is processed.
- Once a face is detected, **facial landmark detection** is applied to extract key points such as eyes, nose, and mouth, which are essential for further analysis.
- These landmarks provide a detailed representation of the face, enabling the system to track **eye movement and head orientation** precisely.
- Efficient models ensure that detection is performed in **real-time**, maintaining system responsiveness.
- This stage is critical because accurate detection directly impacts the **reliability of attention analysis**.

6.3 Attention Analysis Logic

- The system analyses **eye position and head orientation** to determine whether the user is focused on the screen or distracted. These features act as primary indicators of attention.
- Logical conditions are defined to classify attention, such as checking if the user is **looking straight at the screen or frequently looking away**.
- The system may also consider patterns like **blinking frequency and head movement**, which provide additional insights into user behaviour.
- Based on these conditions, the system categorizes the user as **attentive or inattentive** and updates the result continuously.
- This logic ensures **real-time decision-making**, allowing immediate feedback and accurate monitoring.

6.4 GUI Development

- A **Graphical User Interface (GUI)** is developed to provide an interactive platform for users to operate the system easily without technical knowledge.
- The GUI includes features such as **start/stop controls**, real-time video display, and attention status indicators.
- Results are displayed clearly using **text or visual cues**, making it easy for users to understand their attention level instantly.
- The interface is designed to be **simple, clean, and user-friendly**, ensuring smooth interaction.
- This component enhances the overall usability of the system by combining functionality with **ease of access and visualization**.

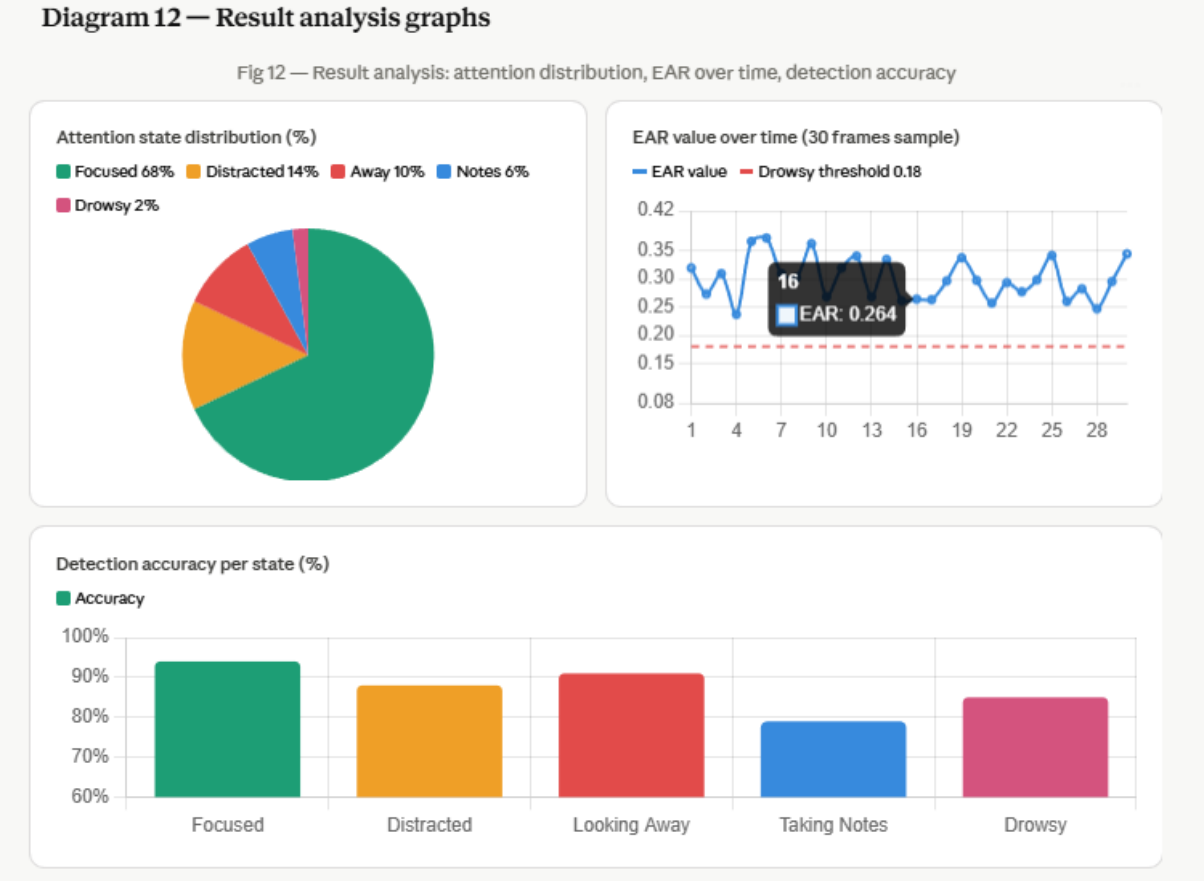
Chapter 7 — Results and Evaluation

7.1 System Output

- The system provides **real-time output** by displaying whether the user is attentive or inattentive based on the analysis of webcam input. The result is updated continuously as new frames are processed.
- Visual indicators such as **on-screen text labels, bounding boxes, or status messages** are used to clearly represent the attention state.
- When the user is looking at the screen with proper head alignment, the system shows **"Attentive"**, while frequent distractions or looking away results in **"Inattentive"** status.
- The output is integrated into the **GUI interface**, allowing users to view both the live video feed and the detection result simultaneously.
- This real-time feedback helps users understand their **focus behaviour instantly** and adjust accordingly.

7.2 Result Analysis

- The system performs effectively under **normal lighting conditions**, where facial features are clearly visible and easy to detect.
- Accuracy may slightly decrease in cases of **poor lighting, occlusions (like hands or objects covering the face), or extreme head angles**, which can affect detection reliability.
- The combination of **face detection, landmark detection, and attention logic** provides a balanced approach, improving overall detection accuracy.
- Continuous frame processing ensures that temporary errors are corrected quickly, maintaining **stable and consistent output over time**.
- The results demonstrate that the system is capable of **reliably distinguishing between attentive and inattentive states** in most practical scenarios.



7.3 Performance Evaluation

- The system achieves **real-time performance**, processing frames quickly without noticeable delay, making it suitable for live monitoring applications.
- Resource usage is optimized, allowing the system to run smoothly on **standard hardware without excessive CPU or memory consumption**.
- The use of efficient libraries ensures **fast detection speed and acceptable accuracy**, even without high-end GPUs.
- Performance is influenced by factors such as **camera quality, lighting conditions, and system specifications**, which can impact detection speed and accuracy.
- Overall, the system provides a good balance between **speed, accuracy, and efficiency**, making it practical for real-world implementation.

Chapter 8 — Conclusion and Future Scope

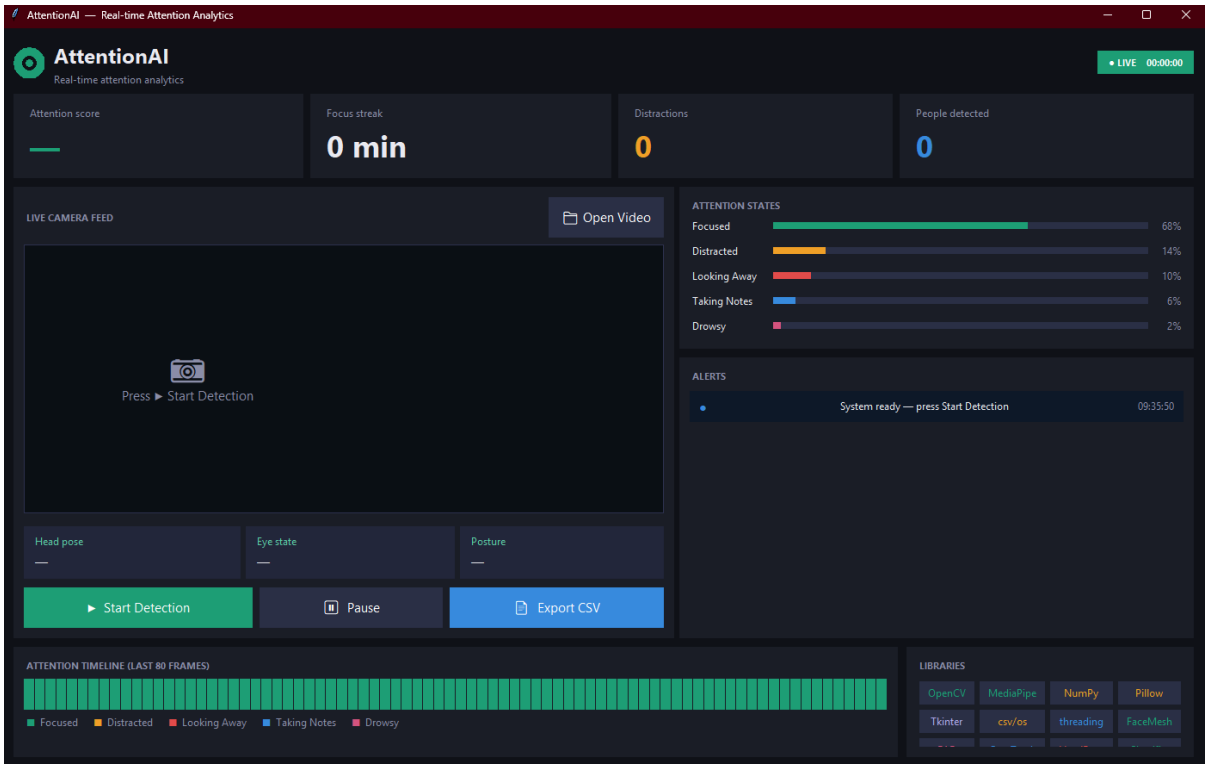
8.1 Conclusion

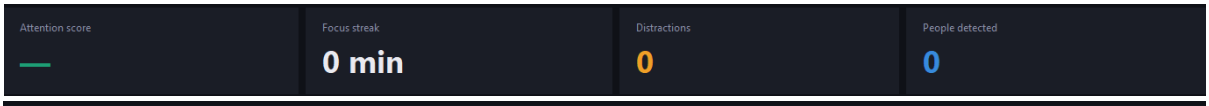
- The project successfully implements an **AI-based attention monitoring system** that can detect user focus in real-time using webcam input and computer vision techniques. It addresses the problem of lack of supervision in online environments.
- The system uses **face detection, facial landmark detection, and attention analysis logic** to classify whether a user is attentive or inattentive with reasonable accuracy.
- It provides a **cost-effective and practical solution**, as it does not require specialized hardware and can run on standard computers.
- The integration of a **simple GUI interface** makes the system easy to use and accessible to non-technical users.
- Overall, the project demonstrates the effective use of **Artificial Intelligence and computer vision** in solving real-world problems related to attention monitoring.

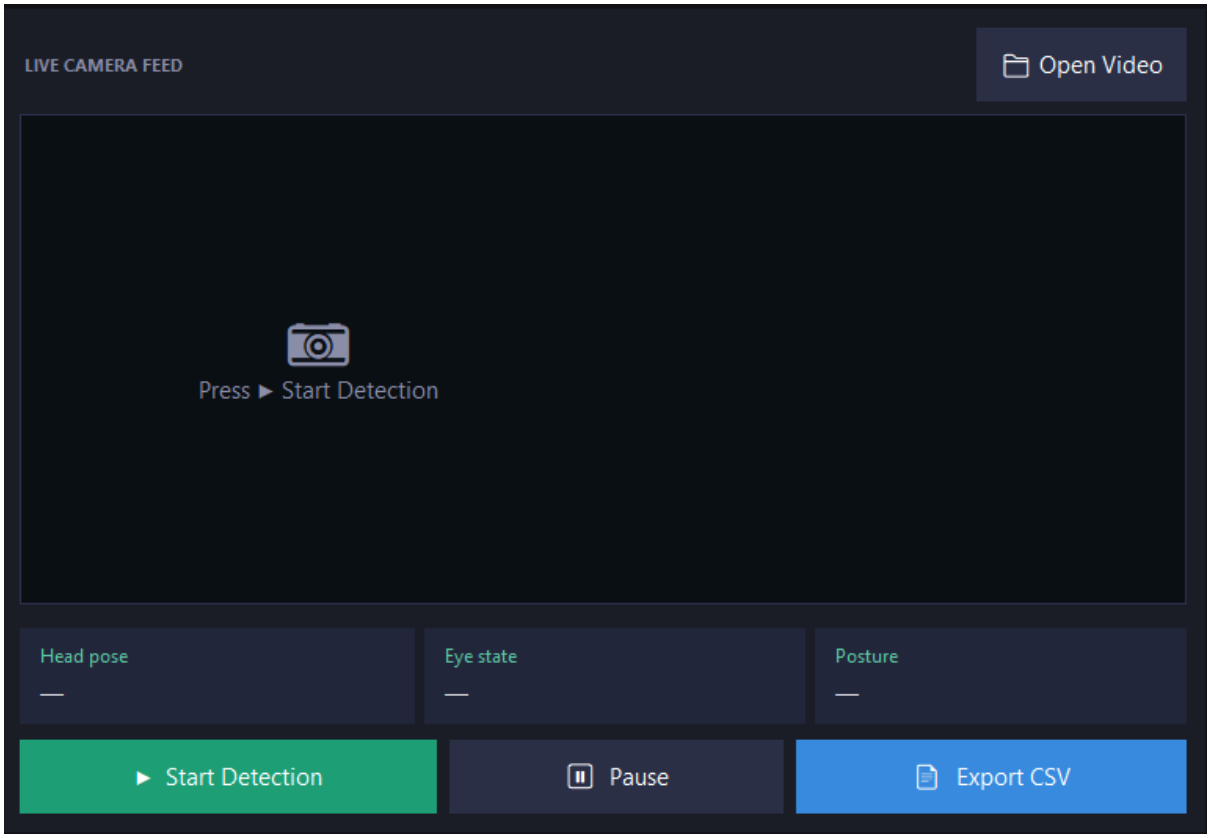
8.2 Future Scope

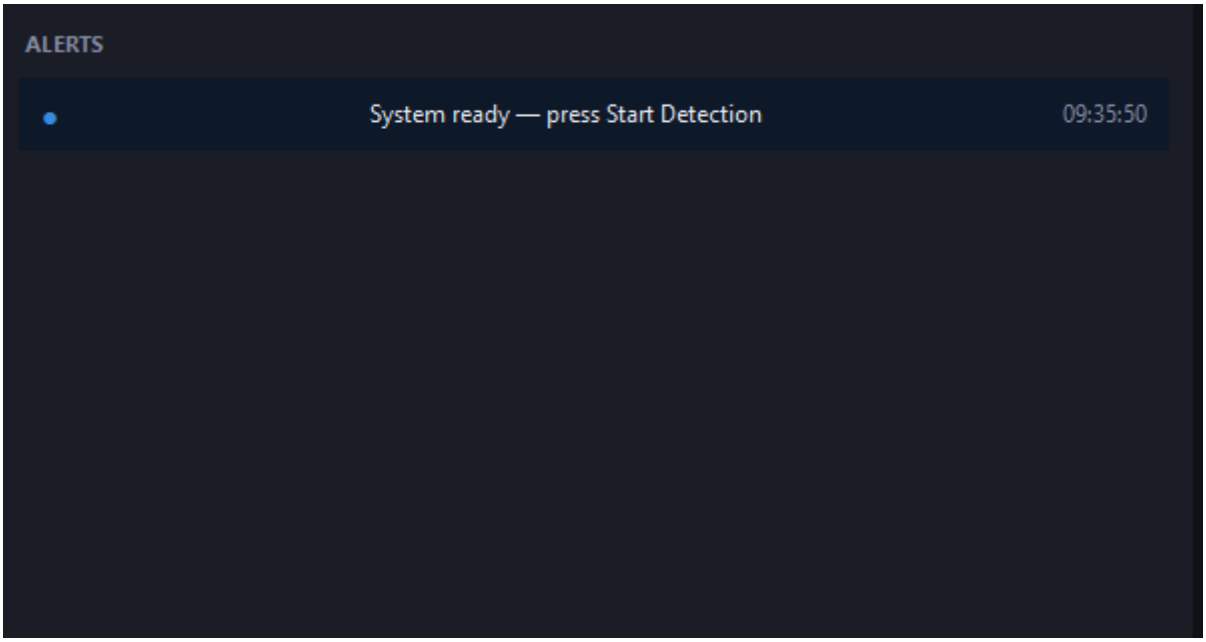
- The system can be enhanced by integrating **advanced deep learning models**, which can improve accuracy and handle more complex scenarios like multiple users or varied environments.
- Additional features such as **emotion detection and fatigue analysis** can be included to provide deeper insights into user behaviour and engagement levels.
- The system can be integrated with **online learning platforms or workplace tools**, making it more useful in real-world applications.
- Support for **multi-user detection** can be added to monitor multiple individuals simultaneously in a shared environment.
- Future improvements can focus on making the system more **robust against lighting variations, occlusions, and different camera qualities**.

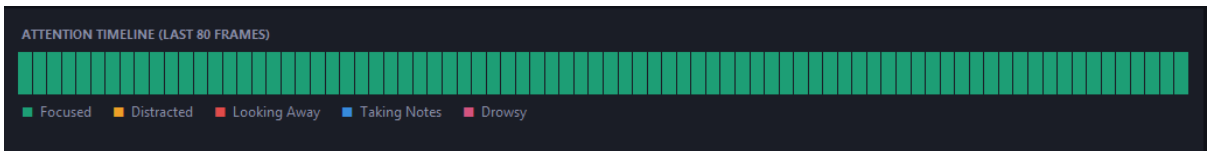
8.3 Screenshots and Observations

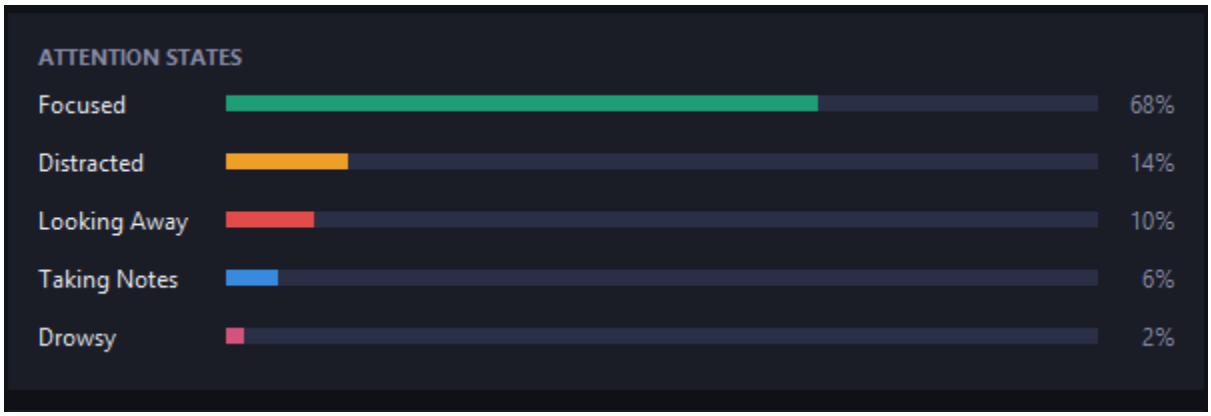


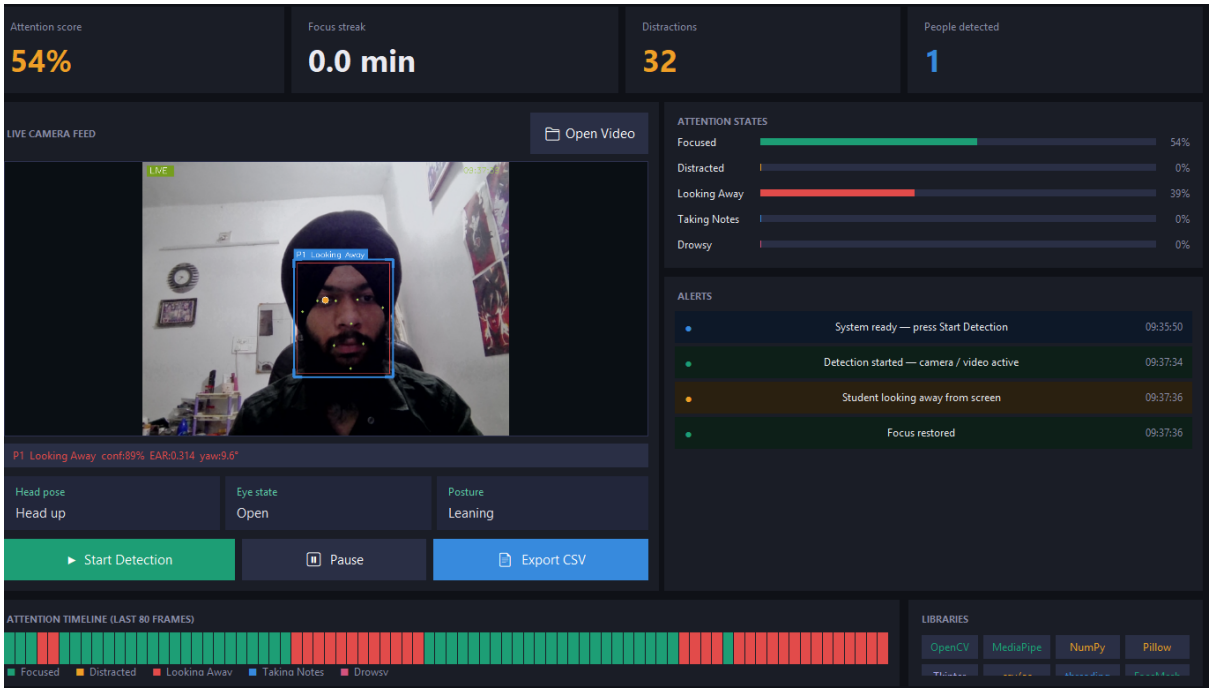


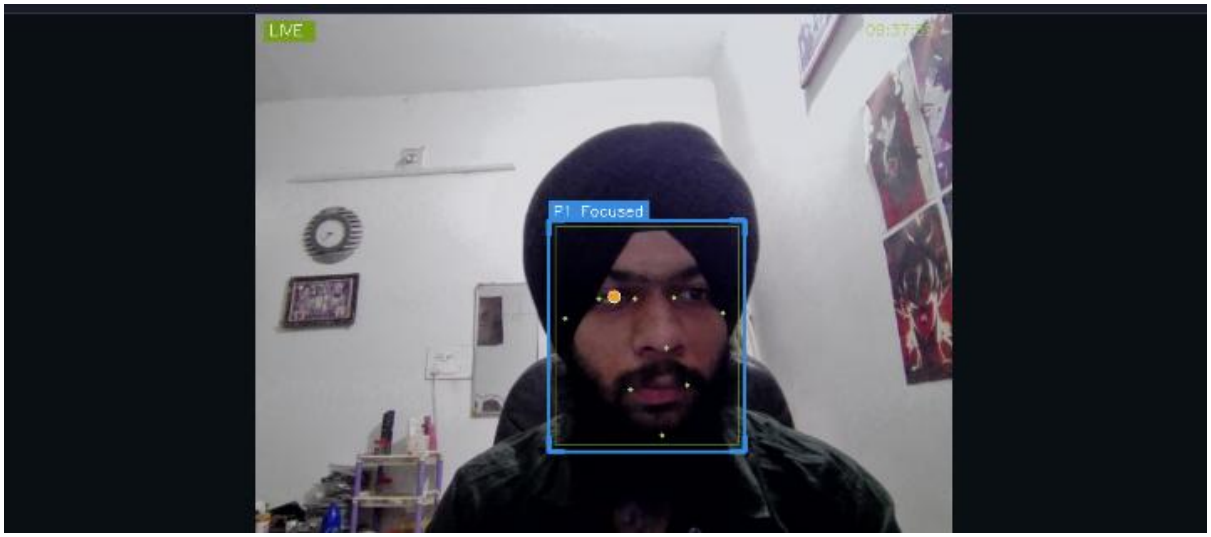


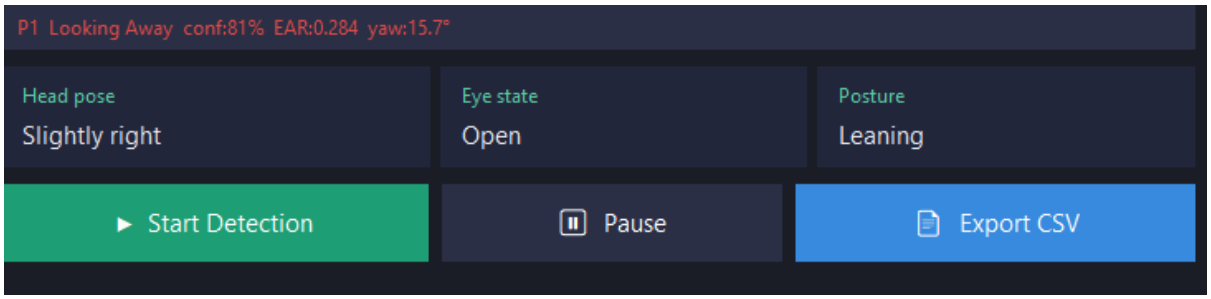

ALERTS
System ready — press Start Detection
09:35:50

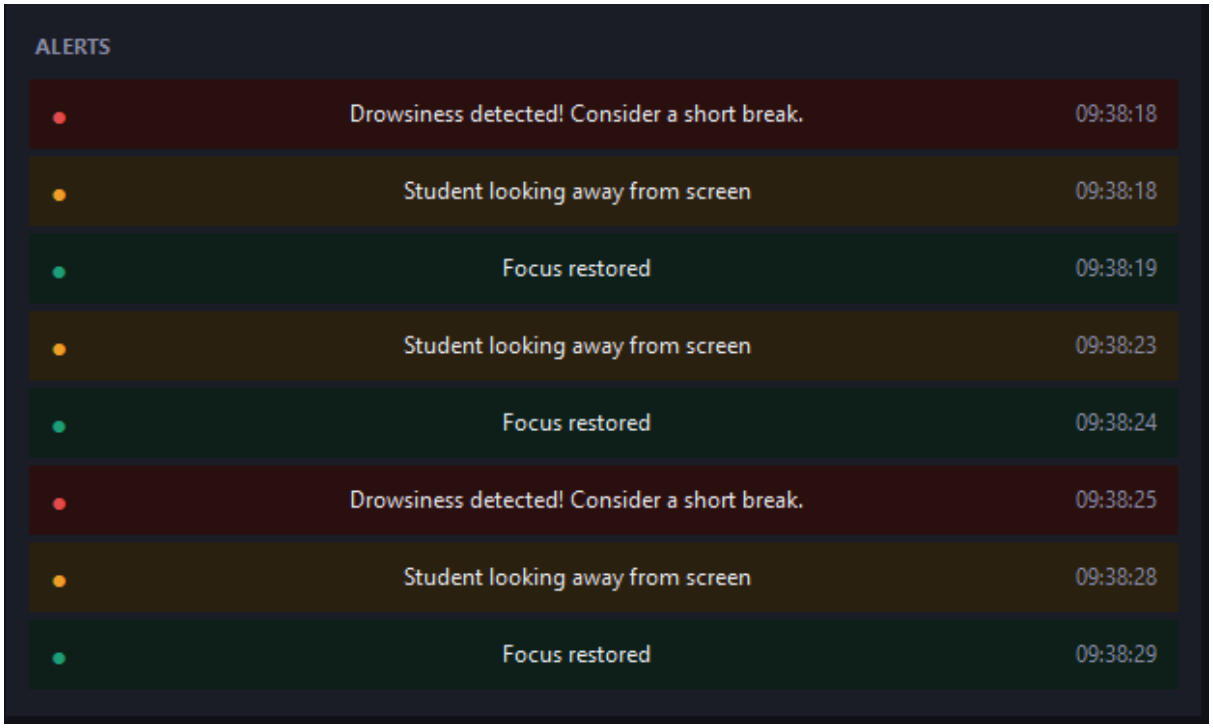

ATTENTION TIMELINE (LAST 80 FRAMES)
Focused
Distracted
Looking Away
Taking Notes
Drowsy

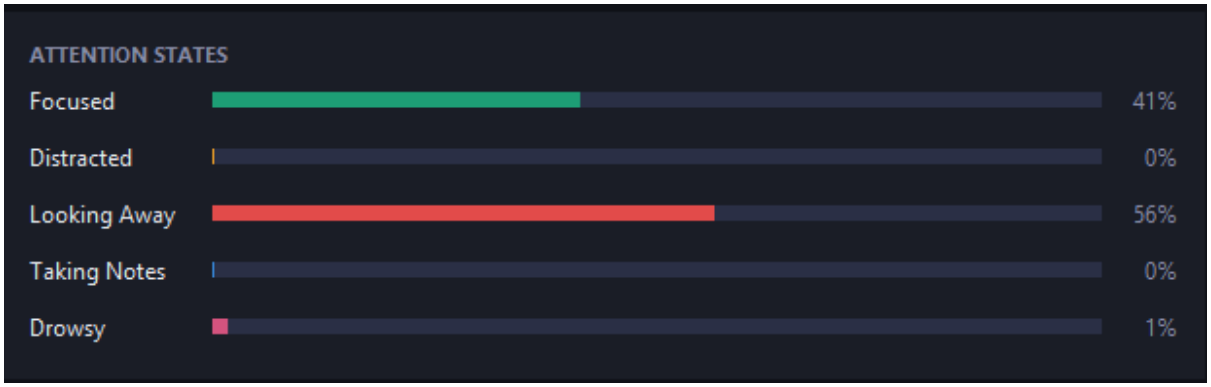

ATTENTION STATES
Focused
68%
Distracted
14%
Looking Away
10%
Taking Notes
6%
Drowsy
2%

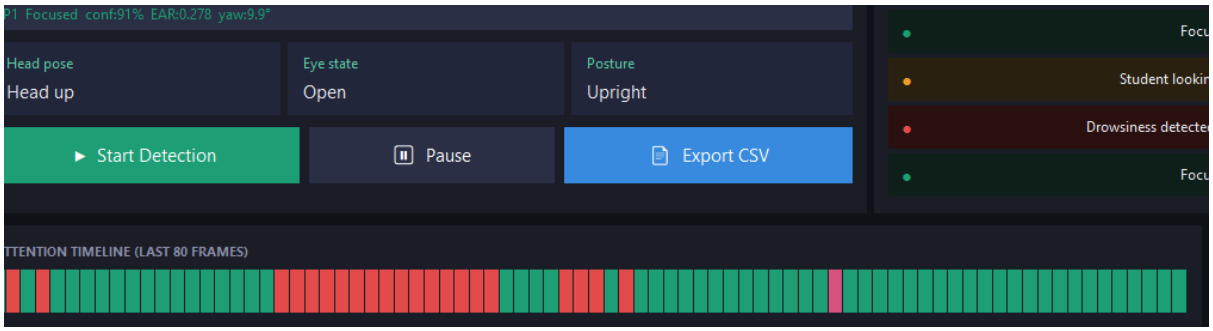

Attention score
54%
Focus streak
0.0 min
Distractions
32
People detected
1
LIVE CAMERA FEED
Open Video
LIVE
P1 Looking Away
P1 Looking Away conf:89% EAR:0.314 yaw:9.6°
Head pose
Head up
Eye state
Open
Posture
Leaning
Start Detection
Pause
Export CSV
ATTENTION STATES
Focused 54%
Distracted 0%
Looking Away 39%
Taking Notes 0%
Drowsy 0%
ALERTS
System ready — press Start Detection 09:35:50
Detection started — camera / video active 09:37:34
Student looking away from screen 09:37:36
Focus restored 09:37:36
ATTENTION TIMELINE (LAST 80 FRAMES)
Focused
Distracted
Looking Away
Taking Notes
Drowsy
LIBRARIES
OpenCV
MediaPipe
NumPy
Pillow


LIVE
P1 Focused


P1 Looking Away conf:81% EAR:0.284 yaw:15.7°
Head pose
Slightly right
Eye state
Open
Posture
Leaning
Start Detection
Pause
Export CSV


ALERTS
Drowsiness detected! Consider a short break. 09:38:18
Student looking away from screen 09:38:18
Focus restored 09:38:19
Student looking away from screen 09:38:23
Focus restored 09:38:24
Drowsiness detected! Consider a short break. 09:38:25
Student looking away from screen 09:38:28
Focus restored 09:38:29


ATTENTION STATES
Focused 41%
Distracted 0%
Looking Away 56%
Taking Notes 0%
Drowsy 1%


Head pose
Head up
Eye state
Open
Posture
Upright
Start Detection
Pause
Export CSV
ATTENTION TIMELINE (LAST 80 FRAMES)

References

- Research papers and articles on **attention detection and computer vision techniques** used for monitoring user [1]behaviour.

  1. Real-Time Eye Tracking / Attention

  Title: *Real-Time Eye Gaze Tracking for Human Computer Interaction*

  Author: Qiang Ji

  2. Facial Landmark Detection (Core of your system)

  Title: *Real-time Facial Surface Geometry from Monocular Video on Mobile GPUs*

  Author: Yury Kartynnik

  3. Deep Learning for Face Detection

  Title: *Deep Learning Face Detection: A Survey*

  Author: Jianjia Li

  4. Head Pose Estimation

  Title: *Head Pose Estimation in Computer Vision: A Survey*

  Author: Erroll Wood

  5. Driver Attention Monitoring (same concept as your project)

  Title: *Driver Attention Monitoring Using Computer Vision*

  Author: Mohamed Abouelnaga

- Documentation of libraries such as **OpenCV and MediaPipe**, which are used for image processing and facial landmark detection.
- Python official documentation for understanding **programming concepts and implementation details**.
- Online resources and tutorials related to **AI-based real-time detection systems and GUI development**.